- Defining the problem

We needed to create a detection system that successfully detects a fire in a short period of time. Our group had the opportunity to talk to a fire protection engineer on what worked and what didn't. Something that did work for us was the light and sound was working for the fire detection. The issue was that the actual detection didn't work meaning water wasn't being distributed.

- Ideation

Our overall idea was to have a fire suppression system at the roof of the house, then have clear tubes collecting the water then releasing it into the fire. This idea was good

- Selection & prototype design

We did research on how to place a fire detection system in our room/apartment building. This made our model go well since it was realistic livable.

- Prototype building

Building our prototype went much easier since we had the structure lay out and measurements given to us but the burn manual

- Testing

During our final test, we used another group's house so the ending result is based on their design. The fire was detected in 5 seconds. The water took a little longer to stop the fire. It took about a Minute and 30 seconds for the water to go out

- Evaluation

Overall, our designs, prototype and detection with the light and sound went well. The only problem that occured was the water was't being left out.

Based on my experience with this project, designers need to first think about what their goal is, how will they achieve that goal, cost of materials, what their prototype will be, and the main idea of their design. A way to improve your design in the future is to look at your old design to see what went well and what went wrong.